Rhetorical Adaptation Following Party Switching: Evidence from Brazil's Chamber of Deputies

Arthur Gomes Nery

January 2026

Abstract

When legislators switch parties, does their rhetoric change? We analyze 365,551 legislative speeches from Brazil's Chamber of Deputies (2003–2025) to measure linguistic adaptation following 1,642 party switches. Using an event study design with Double Machine Learning, we find that deputies' speeches become 3.9 percentage points less likely to be classified as their old party within one bimester of switching ($p < 0.001$), a 34% decline from baseline. A decomposition using named entity recognition indicates that approximately 73% of the detected change reflects shifts in whom deputies reference, while 27% reflects changes in policy vocabulary. We also examine voting behavior: deputies show a marginally significant increase in loyalty to their new party (2.9 percentage points, $p = 0.063$), but among the subset of switchers with sufficient data ($N = 247$), linguistic and voting changes are not correlated. Exploratory heterogeneity analysis indicates that adaptation varies by switch direction (rightward switches show larger effects, $p = 0.025$) but not by career experience or ideological distance. These findings document substantial and immediate rhetorical change following party switches, with political references accounting for the majority of the detected signal.

Keywords: Party switching, legislative behavior, political rhetoric, text-as-data, Brazil, machine learning

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1 Introduction

Party labels serve as informational shortcuts for voters, signaling how representatives are likely to behave in office. In Brazil's Chamber of Deputies, approximately one-quarter of legislators switch parties during each four-year term. This fluidity raises a basic descriptive question: when deputies change their party affiliation, does their behavior change as well? If switching deputies continue to behave as before, party labels carry limited information. If their behavior shifts to match their new party, labels retain meaning despite the instability of individual affiliations. This paper examines whether and how legislative rhetoric changes following party switches.

We analyze 365,551 floor speeches delivered in Brazil's Chamber of Deputies between 2003 and 2025, measuring linguistic change around 1,642 party switches involving 933 deputies. Our approach uses a text classifier trained on non-switching deputies to quantify how closely each speech resembles the rhetorical patterns of different parties. We then apply an event study design with double machine learning to estimate changes in this measure before and after switches. The classifier achieves 28.9% accuracy on a 25-party classification task, exceeding the random baseline by a factor of 7.23. We emphasize that this method tracks party-coded rhetoric and named entity mentions, the linguistic patterns that distinguish parties in floor speeches, rather than underlying ideology or beliefs.

We find substantial linguistic adaptation. Within one bimester of switching, deputies' speeches become 3.9 percentage points less likely to be classified as their former party ($p < 0.001$), representing a 34% decline from baseline. This change is detectable immediately after the switch date and persists through the full observation window of six bimesters. All three estimation approaches we employ (ordinary least squares, two-way fixed effects, and double machine learning) yield similar estimates and pass the parallel trends test, with pre-period coefficients statistically indistinguishable from zero.

A decomposition of the adaptation signal indicates that political references account for the majority of the detected change, with policy vocabulary contributing a smaller but statistically significant component. Using named entity recognition to remove mentions of politicians, organizations, and locations, we estimate the relative contributions of each channel.

We also examine voting behavior. Deputies show a marginally significant increase in voting loyalty to their new party, but among switchers with sufficient data to compute individual-level measures, we find no positive correlation between linguistic and voting changes. This analysis covers only 19.4% of switchers due to data requirements; whether the pattern generalizes to less active deputies is unknown.

We conduct exploratory heterogeneity analysis to examine whether adaptation varies across observable characteristics. Adaptation does not differ significantly by career experience or by the ideological distance of the switch. However, adaptation does vary by switch direction: rightward switches produce effects approximately 60% larger than leftward switches ($p = 0.025$). We interpret these patterns cautiously given the exploratory nature of the analysis and the possibility that the direction finding reflects asymmetries in party structure or classifier performance rather than differences in behavioral adaptation.

The paper proceeds as follows. Section 2 reviews the literature on party switching and legislative behavior. Section 3 presents our conceptual framework. Section 4 describes the data and measurement strategies. Section 5 details the empirical approach. Section 6 reports the main results. Section 7 presents robustness checks. Section 8 examines heterogeneity. Section 9 discusses the findings and their limitations.

2 Related Work

2.1 Party Switching in Brazil

Brazil's party system exhibits high fragmentation and fluidity. As of 2025, the country has 30 registered parties, with 20 or more typically represented in the Chamber of Deputies at any given time (Mainwaring et al., 2018). Most parties are weakly institutionalized, lacking deep organizational roots. Voter attachments are shallow and volatile, and elite behavior is characterized by frequent defection and realignment (Desposato, 2006; Mainwaring, 1999).

This institutional environment reflects Brazil's political development. The current party system emerged after the 1985 democratic transition, providing limited time for parties to develop organizational depth or stable voter loyalties. The electoral system (open-list proportional representation with large district magnitudes) weakens party cohesion by encouraging personalistic campaigns and intraparty competition (Ames, 2001). Executive dominance in the legislative process, combined with extensive patronage resources controlled by the president, creates incentives for deputies to prioritize access to the governing coalition over party loyalty (Figueiredo, 2001). Brazil's coalitional presidentialism requires the formation of multiparty cabinets, with ministerial appointments functioning as a key mechanism of executive-legislative relations (Amorim Neto, 2000).

Party switching is common in this context. Approximately one-quarter of deputies switch at least once per four-year legislature (Desposato, 2006). Some deputies switch multiple times. In our sample, one deputy switched 8 times over the 22-year observation period, and 406 deputies switched more than once. Switching clusters temporally, with rates peaking at legislature midpoints when coalition negotiations intensify and immediately following presidential elections when governing coalitions are reconstituted.

A substantial literature examines the determinants of party switching in Brazil. Studies have identified several motives.

First, access to executive resources plays a role. The Brazilian president controls patronage resources including ministerial appointments, budget amendments, and targeted spending allocations. Joining the government coalition provides access to these resources, creating incentives to switch into coalition parties (Ames, 2001; Amorim Neto, 2000; Pereira & Mueller, 2004). Deputies from opposition parties face disadvantages in delivering benefits to their constituencies.

Second, electoral calculations matter. Deputies switch to parties with better local organization, stronger ballot position, or more favorable coalition prospects in their districts (Desposato, 2006). Under open-list proportional representation, a deputy's electoral outcome depends partly on total votes received by their party list, creating incentives to affiliate with locally stronger parties. Access to campaign resources and media time also varies with party size and coalition status.

Third, coalition realignment drives some switches. When governing coalitions shift, particularly following presidential elections, deputies may switch to remain in or join the majority (Figueiredo & Limongi, 1999). Party leaders sometimes orchestrate collective switches, moving blocs of deputies into new coalition arrangements. The 2016 impeachment of President Dilma Rousseff triggered realignment as deputies repositioned relative to the Temer government.

Fourth, some switches reflect ideological repositioning, particularly in response to major political realignments or the emergence of new parties (Freitas, 2012). The rise of new right-wing parties in the 2010s attracted deputies seeking ideologically aligned affiliations. Resource allocation between federal and subnational governments further shapes switching incentives, as deputies align with parties that control state-level resources (Desposato & Scheiner, 2008).

Fifth, career advancement considerations influence switching. Switches may facilitate leadership positions, committee assignments, or future candidacies that would be blocked in a deputy's current party (O'brien & Shomer, 2013). Junior deputies facing limited advancement opportunities may find better prospects in parties with fewer senior members.

This literature treats the switch itself as the outcome to be explained. Less attention has been devoted to what happens to deputies' behavior after they change parties.

2.2 Legislative Behavior After Switching

A smaller literature examines behavioral change following party switches. Heller and Mershon (2005) study party switching in Italy, finding that switching is most common when party labels provide limited policy information and pit copartisans against each other in serving constituent needs. Their subsequent voting patterns shift toward their new party but not completely. Desposato (2006) examines roll-call voting in Brazil, finding that switchers' voting patterns shift partially but not completely toward their new party. Nokken (2000) studies party switchers in the United States, documenting voting changes that increase alignment with the new party. Nokken and Poole (2004) extend this to a historical analysis, finding that the largest behavioral changes occur during periods of high ideological polarization.

These studies focus on roll-call voting. Voting is consequential and readily measurable, but it captures one dimension of legislative behavior. Votes are public, strategic, and constrained by party discipline. Legislative speech offers a different window into legislative behavior, capturing how deputies frame issues, what language they use, and how they position themselves rhetorically.

2.3 Computational Analysis of Political Text

Advances in computational text analysis have enabled systematic study of legislative speech (Grimmer & Stewart, 2013). Researchers have used text analysis to estimate legislator ideology (Slapin & Proksch, 2008), measure partisan rhetoric (Gentzkow et al., 2019), and detect policy positions (Laver et al., 2003). This literature has shown that speech contains information about political positioning beyond what roll-call votes capture.

Early approaches to extracting positions from text include the "Wordscores" method, which uses reference texts to score new documents (Laver et al., 2003), and "Wordfish," which estimates positions as latent variables from word frequencies without requiring reference texts (Slapin & Proksch, 2008). These methods have been applied extensively to party manifestos and legislative speeches across multiple countries.

More recent work has developed sophisticated methods for measuring group differences in text. Gentzkow et al. (2019) address finite-sample bias in measuring partisan differences in congressional speech, finding that partisanship increased sharply in the early 1990s after remaining low and relatively constant over the preceding century. Their approach defines partisanship as the ease with which an observer could infer a legislator's party from a single utterance, a conceptualization that aligns with our use of text classifiers to measure rhetorical distinctiveness.

In the Brazilian context, Figueredo et al. (2022) apply supervised classification to over

85,000 Senate speeches spanning 1995–2018, using the Power and Zucco expert survey as training labels. They find that classification accuracy (interpreted as a measure of polarization following Peterson and Spirling (2018)) increases steadily across legislatures, particularly after the PT assumed the presidency in 2003. Their feature analysis reveals that left-wing senators emphasize terms related to social issues (land reform, rights, women, children), while right-wing senators focus on production, agriculture, and religious themes. Notably, when validating their results with the Wordfish scaling algorithm, they find discrepancies that suggest government-opposition dynamics may contaminate left-right classifications, consistent with earlier findings on roll-call-based measures.

Topic models have been used to identify the substantive content of political speech and track its evolution over time (Blei et al., 2003; Roberts et al., 2019). These methods complement position-scaling approaches by revealing *what* legislators discuss rather than *where* they stand.

We extend this work by examining linguistic change following party switches. We train a classifier to predict party affiliation from speech and use the classifier output to measure how closely each speech resembles different parties' rhetorical patterns. This approach allows us to examine whether deputies' speech shifts following a switch and to decompose the sources of any detected change. We also explore whether linguistic changes are associated with changes in voting behavior at the individual level.

Our study contributes to the intersection of three literatures: the Brazilian politics literature on party switching determinants, the comparative politics literature on post-switching behavioral change, and the computational social science literature on measuring political positions from text. By combining causal inference methods with text classification, we provide new evidence on the nature and sources of rhetorical adaptation following party switching.

3 Conceptual Framework

This section outlines the conceptual basis for our empirical analysis. We describe reasons why adaptation might occur following party switches, the dimensions of adaptation we measure, and the heterogeneity patterns we examine.

3.1 Drivers of Adaptation

Several processes could produce behavioral change following a party switch. Deputies may change their beliefs over time, with the switch reflecting a genuine shift in political orientation. Deputies may adjust their behavior strategically to establish credibility with new colleagues and gain access to party resources. Deputies may absorb the speech patterns of their new environment through routine social interaction without deliberate intent. These processes are not mutually exclusive, and our research design does not allow us to distinguish among them directly.

We note that the Brazilian context, where switching is frequent and often tracks coalition realignments, suggests that strategic considerations play a substantial role in switching decisions. However, the motivation for switching need not determine the mechanism of subsequent adaptation. A deputy who switches for strategic reasons might nonetheless absorb new rhetorical patterns through social exposure.

Rather than testing competing theories of adaptation, we focus on documenting whether adaptation occurs, characterizing its magnitude and timing, and decomposing its sources. The patterns we observe may be more or less consistent with different accounts, but we do not claim to identify the underlying mechanism.

3.2 Dimensions of Adaptation

We measure adaptation along two dimensions: legislative speech and voting behavior.

For speech, we use a text classifier to measure how closely each speech resembles the rhetorical patterns of different parties. Our outcome variable is the probability that a classifier, trained on non-switching deputies, assigns a speech to the deputy's former party. A decline in this probability following a switch indicates that the deputy's rhetoric has moved away from their old party's linguistic signature.

We further decompose linguistic adaptation into two components. The first component reflects changes in whom deputies reference: the politicians, organizations, and locations mentioned in speeches. The second component reflects changes in policy vocabulary after removing all named entities. This decomposition distinguishes between changes in political references and changes in substantive policy language.

For voting, we measure loyalty to the party majority position before and after switching. We also examine whether linguistic and voting changes are correlated at the individual level. A positive correlation would suggest that both forms of adaptation reflect a common underlying process. The absence of correlation would suggest that rhetoric and voting respond to different pressures.

3.3 Heterogeneity

We conduct exploratory analysis of whether adaptation varies across observable characteristics. We examine four dimensions.

First, we examine career experience. Deputies with longer tenures have more established reputations and rhetorical styles. Whether experienced deputies adapt more, less, or equally compared to junior deputies is an empirical question.

Second, we examine ideological distance. Switches that traverse larger distances on the left-right spectrum involve greater repositioning. Whether such switches produce larger or smaller adaptation effects is unclear a priori; larger distances might require more adjustment, or might involve deputies who are less committed to any particular partisan identity.

Third, we examine switch direction. Rightward and leftward switches occur in different political contexts and may involve parties with different organizational characteristics or rhetorical distinctiveness. Asymmetries by direction could reflect differences in party structure, differences in the types of deputies who switch in each direction, or differences in classifier performance across parties.

Finally, we examine heterogeneity by destination party size. Larger parties may be more heterogeneous coalitions with less distinctive linguistic signatures, making adaptation harder to detect or less necessary for integration. Deputies who select into minor parties may also differ from those joining major parties on unobserved characteristics related to ideological commitment or rhetorical flexibility. Whether adaptation is larger or smaller for switches to minor versus major parties is thus an empirical question with plausible arguments in either direction.

We treat these analyses as exploratory. We did not pre-register directional hypotheses, and we examine multiple dimensions, raising concerns about multiple comparisons. Significant heterogeneity may be informative, but null results are ambiguous: they could reflect genuine uniformity or insufficient statistical power to detect moderate differences.

4 Data and Measurement

This section describes our data sources and measurement strategies. We draw on three primary sources: legislative speeches from the Chamber of Deputies, party switching records from deputy affiliation histories, and roll-call voting records. We also describe the construction of control variables and topic measures.

4.1 Speech Corpus

Our primary data source is a corpus of legislative speeches from the Brazilian Chamber of Deputies, accessed through the Chamber's Open Data API.[1] The corpus covers 7 four-year

[1] https://dadosabertos.camara.leg.br/

legislatures (52nd through 58th, spanning 2003 to 2025) and comprises 365,551 speeches delivered by 1,634 unique deputies.

The raw transcripts contain noise that must be addressed before analysis. Parliamentary speeches include procedural boilerplate, attached documents, and transcription artifacts. We implement a multi-level preprocessing pipeline:

- **Level 1** performs structural cleaning. We remove parliamentary boilerplate phrases that appear formulaically across speeches, extract the primary speaker's text while removing quoted material and attachments, filter speeches shorter than 100 characters, and remove exact duplicates.
- **Level 2** performs normalization. We convert text to lowercase and remove punctuation and numeric characters. We remove high-frequency, low-information phrases identified using inverse document frequency scores, Portuguese stopwords, and domain-specific irrelevant terms.
- **Level 3** removes named entities. We apply spaCy's Portuguese named entity recognition model (Honnibal et al., 2020) to identify and remove person names, locations, and organizations. This level isolates policy content by removing mentions of party leaders, allied politicians, and affiliated organizations.
- **Level 4** applies stemming. We use the RSLP Portuguese stemmer (Orengo & Huyck, 2001) to reduce words to morphological roots, normalizing verb conjugations, noun plurals, and adjectival forms. This level is used for topic modeling.

We use Levels 2 and 3 for the primary analysis, given named entities are part of partisan rhetoric, as whom a deputy mentions reflects their political associations. We use both to decompose our effect between general rhetoric and named entity mentions.

4.2 Party Switches

We identify party switches from deputy affiliation histories in the Chamber's administrative data. The API provides a timeline of each deputy's party affiliations, including the date and nature of each status change. We identify switch events as records where the status description indicates "Alteração de partido" (party change) and the party ID differs between consecutive records.

This procedure yields 1,642 switch events across 933 unique deputies. The distribution is skewed: 527 deputies switched once, while 406 switched twice or more. The maximum is 8 switches by a single deputy over the 22-year period. We treat each switch as a separate event and cluster standard errors at the deputy level to account for within-deputy correlation.

Switching may exhibit temporal clustering, with rates peaking at legislature midpoints when coalition negotiations intensify and immediately following presidential elections when governing coalitions are reconstituted. We address potential confounding with calendar time through legislature fixed effects in the main specification and calendar-month fixed effects in the two-way fixed effects robustness specification.

Figure 5 in the Appendix presents the sample construction and exclusions applied for different analyses.

4.3 Ideology Measures

We encode party ideology using the Power and Zucco expert survey estimates (Power & Rodrigues-Silveira, 2019; Power & Zucco Jr., 2009; Zucco Jr., 2011). The surveys provide continuous estimates ranging from approximately -1 (far left) to $+1$ (far right) and categorical classifications into Left, Center, and Right blocs. We match each party-year to the temporally closest survey estimate. Deputies from parties not covered by the surveys are excluded.

We construct three measures from these data. Ideological distance is the absolute difference in continuous scores between origin and destination parties: $|EST_{\text{new}} - EST_{\text{old}}|$. Direction is coded as rightward if $EST_{\text{new}} > EST_{\text{old}}$ and leftward otherwise. Bloc transition is the categorical transition type, such as Left to Center or Right to Right.

Of 1,642 switches, 1,101 have valid ideology data for both origin and destination parties. The mean ideological distance is 0.259 on the $[-0.96, 0.80]$ scale, indicating that the average switch spans 14.7% of the observed range. 64.8% of switches remain within the same ideological bloc, while 35.2% cross bloc boundaries. Table 27 in the Appendix presents the transition matrix. The distribution is asymmetric: Right-to-Right switches account for 538 events (48.9% of switches with valid ideology), reflecting the preponderance of right-wing parties and the rightward movement of governing coalitions during much of the observation period. Left-to-Left switches are far less common (44 events, 4.0%).

4.4 Voting Data

We obtain roll-call voting records from the Chamber API, comprising 388,911 individual votes. Each record identifies the deputy, vote date, bill or motion, and the deputy's vote (Yes, No, Abstain, or Absent).

For each roll-call vote, we determine the majority position of each party by computing the modal vote among participating members, requiring at least 3 participants for a valid party position. We code each deputy's vote as aligned or misaligned with their party's majority.

We compute party loyalty in symmetric windows around each switch:

$$\text{Loyalty}_{i,p,t} = \frac{\text{Number of votes aligned with party } p \text{ majority in window } t}{\text{Total votes cast by deputy } i \text{ in window } t} \quad (1)$$

For each switcher, we compute four measures: loyalty to the old party in the 180 days before the switch; loyalty to the old party in the 180 days after the switch, representing continued alignment with the former party; loyalty to the new party in the 180 days after the switch; and the change in alignment, computed as post-switch loyalty to the new party minus pre-switch loyalty to the old party.

We restrict the voting analysis to switchers with at least 3 votes in each window, yielding 321 switchers. The correlation analysis in Section 6.3 requires both sufficient votes and sufficient speeches, yielding 247 switchers (Figure 5).

4.5 Controls

We condition on time-varying factors that may influence both switching propensity and linguistic patterns.

- **Career tenure** records years since a deputy's first election to the Chamber. Experienced deputies may have more established rhetorical patterns. Tenure may also correlate with switching, as junior and senior deputies face different incentives.

- **Legislative activity** measures formal participation as the annual count of bills and amendments proposed by each deputy. Activity may correlate with both speech production and switching propensity.

- **Legislature fixed effects** absorb common shocks affecting all deputies within each four-year term, including policy agendas, major political events, and trends in political discourse.

- **Calendar-month fixed effects**, included in the two-way fixed effects specification, absorb month-specific shocks such as budget debates or electoral periods. The main specification includes legislature fixed effects and topic controls but not calendar-month effects.

4.6 Topic Model

A concern is that the classifier might detect changes in what deputies discuss rather than how they discuss it. If deputies receive different committee assignments after switching and speak about different policy domains, the classifier might detect subject-matter shifts. We address this by controlling for speech topics using Latent Dirichlet Allocation (LDA).

LDA represents each document as a mixture of latent topics, where each topic is a distribution over words (Blei et al., 2003). The model assumes the following generative process for each document d: draw topic proportions $\boldsymbol{\theta}_d \sim \text{Dirichlet}(\boldsymbol{\alpha})$; for each word position, draw a topic assignment from $\boldsymbol{\theta}_d$ and then draw a word from the topic's word distribution. The joint probability of words and topic assignments is:

$$p(\mathbf{w}_d, \mathbf{z}_d, \boldsymbol{\theta}_d | \boldsymbol{\alpha}, \boldsymbol{\Phi}) = p(\boldsymbol{\theta}_d | \boldsymbol{\alpha}) \prod_{n=1}^{N_d} p(z_{dn} | \boldsymbol{\theta}_d) p(w_{dn} | \boldsymbol{\phi}_{z_{dn}}) \tag{2}$$

We estimate the model using variational Bayes on the full corpus with Level 4 stemmed text. We select $K = 20$ topics based on coherence scores across $K \in \{5, 10, \ldots, 50\}$. The document-term matrix uses maximum document frequency of 0.30 and minimum frequency of 300. Table 26 in the Appendix presents the top words for each topic.

For each speech, we identify the dominant topic $\hat{z}_d = \arg\max_k \hat{\theta}_{dk}$ and include it as a categorical control using 19 indicator variables. This conditions on subject matter, so estimated effects reflect rhetorical changes within topics rather than shifts across topics.

Topic choice may be endogenous to switching if deputies shift their policy focus after joining a new party. Conditioning on topics could therefore absorb part of the effect. This makes our estimates conservative: detected effects represent adaptation within topics.

5 Empirical Strategy

Our approach combines three elements: a text classifier trained on non-switching deputies to measure partisan rhetoric, an event study design to estimate changes around switching events, and double machine learning to control flexibly for observed confounders. This section describes each element.

5.1 Party Classifier

5.1.1 Data Partitioning

Our measurement strategy requires a stable benchmark against which to assess whether switchers' speeches become more or less aligned with their old party. Training a classifier on all speeches would be problematic because it would incorporate switchers' post-switch speeches, potentially contaminating the benchmark with the adaptation we seek to measure.

We address this by partitioning deputies into three groups. The training set consists of 80% of deputies who never switched during our observation period (611 deputies, 139,677 speeches). These deputies provide a benchmark of party rhetoric uncontaminated by switching. The test set consists of all switching deputies who appear in the speech corpus (871 deputies, 186,082 speeches).[2] The holdout set consists of the remaining 20% of non-switchers (152 deputies, 39,792 speeches), reserved for out-of-sample diagnostics.

This partition ensures that the classifier has not been trained on adaptation. When applied to switchers' speeches, detected shifts in classification probabilities reflect changes relative to the stable benchmark of non-switcher rhetoric.

5.1.2 Text Representation

We represent each speech as a numerical vector using Term Frequency-Inverse Document Frequency (TF-IDF). For word w in speech d, the TF-IDF score is:

$$\text{TF-IDF}(w, d) = \underbrace{\frac{f_{w,d}}{\sum_{w' \in d} f_{w',d}}}_{\text{Term Frequency (TF)}} \times \underbrace{\log\left(\frac{N}{n_w}\right)}_{\text{Inverse Document Frequency (IDF)}} \tag{3}$$

where $f_{w,d}$ is the count of word w in document d, N is the total number of documents, and n_w is the number of documents containing word w. The TF component measures word frequency within a speech. The IDF component downweights words appearing in many speeches while upweighting distinctive words.

We retain the 5,000 most frequent terms, including unigrams and bigrams. We exclude terms appearing in fewer than 5 speeches or in more than 70% of the corpus. The resulting matrix has dimensionality $N \times 5{,}000$.

[2] Our administrative records identify 933 deputies with at least one switch event, but 62 of these lack speeches in the cleaned corpus.

5.1.3 Classification Model

We train a multinomial logistic regression classifier to predict party affiliation from TF-IDF vectors. For each party $k \in \{1, \ldots, K\}$ (where $K = 25$), the model learns a weight vector $\boldsymbol{\beta}_k \in \mathbb{R}^{5,000}$. The predicted probability that speech $\mathbf{x}$ belongs to party k is:

$$P(Y = k|\mathbf{x}) = \frac{\exp(\boldsymbol{\beta}_k^\top \mathbf{x})}{\sum_{j=1}^{K} \exp(\boldsymbol{\beta}_j^\top \mathbf{x})} \tag{4}$$

We make two modifications to standard logistic regression.

First, we apply class balancing. Brazilian parties vary in size, from PT with approximately 41% of training speeches to minor parties with less than 1%. Without adjustment, the classifier would achieve high accuracy by predicting PT for every speech. We use class-balanced weights that upweight minority parties inversely proportional to their frequency.

This choice affects interpretation. Our outcome measures adaptation conditional on parties being linguistically distinguishable. If two parties have similar rhetoric, the classifier cannot detect switches between them, and such switches contribute noise to our estimates. Our findings characterize adaptation among switches where origin and destination parties have detectably different linguistic profiles.

Second, we apply L2 regularization. With 5,000 features and 25 classes, overfitting is a concern. We use ridge regularization with penalty parameter $C = 1.0$, selected via cross-validation.

We evaluate classifier performance using both cross-validation on the training set and out-of-sample evaluation on the holdout. Table 1 reports results. Speech-level accuracy is 28.9% under 5-fold cross-validation, exceeding the random baseline of 4.0% by a factor of 7.23, and 26.5% on the holdout. Aggregating predictions within deputy increases accuracy: deputy-level majority-vote accuracy on the holdout is 52.0%.

5.1.4 Outcome Variable

For each speech s delivered by deputy i at time t, the classifier outputs a K-dimensional probability vector:

$$\mathbf{P}(s) = [P(\text{Party}_1|s), P(\text{Party}_2|s), \ldots, P(\text{Party}_K|s)] \tag{5}$$

where $\sum_{k=1}^{K} P(\text{Party}_k|s) = 1$. We extract the probability assigned to the deputy's old party:

$$Y_{it} = P(\text{old party}_i|\text{speech}_{it}) \tag{6}$$

Table 1: Classifier Performance

Dataset	Accuracy	Weighted F1	Speeches	Deputies
Training (5-fold CV)	28.9%	0.326	139,677	611
Holdout (out-of-sample)	26.5%	0.319	39,792	152

Benchmark	Value
Baseline accuracy (uniform random, 25 parties)	4.0%
Improvement over baseline (Training)	7.23×
Improvement over baseline (Holdout)	6.63×

Notes: Accuracy and weighted F1 computed at the speech level. Cross-validation uses GroupKFold by deputy. The holdout consists of non-switcher deputies excluded from training. Deputy-level accuracy assigns each deputy the modal predicted party across speeches.

This variable measures how strongly each speech exhibits the linguistic patterns of the deputy's former party. If adaptation occurs, Y_{it} should decline after switching as speeches become less likely to be classified as the old party.

Using continuous probabilities rather than discrete predictions captures gradations in partisan alignment, provides more statistical power, and avoids threshold effects from predicted class labels.

5.2 Event Study Design

We use an event study framework to trace linguistic change around switching events. For each switch, we define event time as the number of days between the speech date and the switch date. We discretize event time into 60-day bins spanning ±360 days before and after the switch:

$$\tau_{it} = \text{bin}\left(\frac{\text{speech date}_{it} - \text{switch date}_i}{60}\right) \in \{-6, -5, \ldots, -1, +1, \ldots, +6\} \qquad (7)$$

We exclude period 0 (the 60 days immediately surrounding the switch) to avoid ambiguity about whether speeches near the switch date reflect pre- or post-switch behavior. Period $\tau = -1$ serves as the reference category.

The analysis sample includes switch events with at least one speech in the ±360-day window, yielding 47,019 speech-events from 691 unique switchers.

The estimating equation is:

$$Y_{it} = \alpha + \sum_{\tau \neq -1} \beta_\tau \cdot \mathbb{1}[\text{EventTime}_{it} = \tau] + X'_{it}\gamma + \varepsilon_{it} \tag{8}$$

where Y_{it} is the old-party classification probability, $\mathbb{1}[\text{EventTime}_{it} = \tau]$ indicates event time period τ, X_{it} is a vector of controls (career tenure, legislative activity, legislature indicators, topic indicators), and ε_{it} is the error term.

The coefficients $\{\beta_\tau\}$ trace old-party alignment relative to the reference period. Pre-switch coefficients $(\beta_{-6}, \ldots, \beta_{-2})$ test for pre-trends. Post-switch coefficients $(\beta_{+1}, \ldots, \beta_{+6})$ estimate the change in old-party alignment following the switch.

Interpretation of the event study coefficients rests on a counterfactual stability assumption: absent the switch, a deputy's classification probability would remain on the trajectory observed in pre-switch periods, conditional on controls. Non-switchers are used only to train the classifier, not as an outcome control group.

This assumption implies that pre-switch coefficients should be statistically indistinguishable from zero. We assess this using individual coefficient tests for each $\tau \in \{-6, \ldots, -2\}$ and a joint F-test of $H_0 : \beta_{-6} = \beta_{-5} = \beta_{-4} = \beta_{-3} = \beta_{-2} = 0$.

We further validate this assumption through a placebo test. We assign pseudo switch dates to 115 non-switching deputies, drawn from the empirical distribution of actual switch dates, and estimate the same event study specification. If our effects were driven by classifier artifacts or temporal patterns unrelated to actual switching, we would observe similar effects for these placebo events.

We compute cluster-robust standard errors at the deputy level, accounting for serial correlation in speech patterns and correlation across multiple switches by the same deputy.

5.3 Double Machine Learning

Confounding by observed covariates is a concern. Deputies who switch may differ from non-switchers on characteristics that affect speech patterns. Standard linear regression controls impose functional form assumptions; if the true relationship between covariates and outcomes is nonlinear, misspecification can bias treatment effect estimates.

We use double machine learning (DML) (Chernozhukov et al., 2018), which allows flexible modeling of covariate relationships while preserving valid inference on treatment effects.

Estimation Procedure. DML proceeds in three steps. In the first step, we model the conditional expectation of Y given controls X using gradient boosting:

$$\hat{m}(X_{it}) = \hat{E}[Y_{it}|X_{it}] \tag{9}$$

We use a histogram-based gradient boosting regressor with 100 trees and maximum depth 5. We compute outcome residuals $\tilde{Y}_{it} = Y_{it} - \hat{m}(X_{it})$.

In the second step, for each treatment indicator D_{it}^{τ}, we model the propensity score using gradient boosting:

$$\hat{g}_{\tau}(X_{it}) = \hat{P}[D_{it}^{\tau} = 1 | X_{it}] \tag{10}$$

We compute treatment residuals $\tilde{D}_{it}^{\tau} = D_{it}^{\tau} - \hat{g}_{\tau}(X_{it})$.

In the third step, we regress outcome residuals on treatment residuals:

$$\tilde{Y}_{it} = \sum_{\tau} \beta_{\tau} \tilde{D}_{it}^{\tau} + \epsilon_{it} \tag{11}$$

The coefficients $\{\hat{\beta}_{\tau}\}$ are the DML estimates.

Cross-Fitting. Using the same data to train first-stage models and compute residuals would bias residuals toward zero. We implement 3-fold cross-fitting: for each fold, we train $\hat{m}$ and $\{\hat{g}_{\tau}\}$ on the other two folds, compute residuals for observations in the held-out fold, and combine residuals across folds for the final regression. This ensures residuals are computed out-of-sample.

Properties. DML estimates are $\sqrt{N}$-consistent, asymptotically normal, and robust to first-stage misspecification provided first-stage models converge faster than $N^{-1/4}$ (Chernozhukov et al., 2018).

One limitation is that DML may absorb genuine pre-treatment variation along with confounding. The method assumes that systematic covariate-outcome relationships reflect confounding rather than causal dynamics. In our setting, deputies may adapt before the formal switch date, and DML would treat such anticipatory adaptation as confounding. We present results from OLS and TWFE specifications alongside DML to assess sensitivity to this assumption.

6 Main Results

This section presents our main findings. We first document linguistic adaptation following party switches, then examine voting behavior and its relationship to linguistic change.

6.1 Linguistic Adaptation

Table 2 presents the DML event study coefficients for actual switchers alongside placebo estimates for non-switchers, and Figure 1 plots the trajectory of old-party classification probability around switching events.

The event study coefficients β_τ report changes relative to the reference period $\tau = -1$. The window-based average treatment effect in Table 3 reports the mean post-switch minus mean pre-switch difference. The mean reference-period probability is 0.1127, so the peak effect of -0.0387 corresponds to a 34.3% decline from baseline. The placebo test confirms these effects are specific to actual party switching: non-switchers assigned pseudo switch dates show no systematic change (joint post-period test $p = 0.223$).

Table 2: Event Study: Main Results

	Actual Switchers		**Placebo**	
Period	Coefficient	SE	Coefficient	SE
Pre-switch periods:				
$\tau = -6$	+0.0039	0.0098	+0.0147	0.0165
$\tau = -5$	−0.0046	0.0094	+0.0052	0.0175
$\tau = -4$	−0.0059	0.0075	−0.0042	0.0152
$\tau = -3$	−0.0009	0.0074	+0.0043	0.0143
$\tau = -2$	−0.0003	0.0062	+0.0228	0.0128
$\tau = -1$	0.000	—	0.000	—
Post-switch periods:				
$\tau = +1$	-0.0387^{***}	0.0073	+0.0199	0.0112
$\tau = +2$	-0.0390^{***}	0.0083	+0.0078	0.0151
$\tau = +3$	-0.0344^{***}	0.0082	+0.0138	0.0117
$\tau = +4$	-0.0409^{***}	0.0087	+0.0252	0.0134
$\tau = +5$	-0.0419^{***}	0.0090	+0.0017	0.0142
$\tau = +6$	-0.0393^{***}	0.0097	+0.0154	0.0167
Model statistics:				
N (speech-events)	47,019		8,751	
N (deputies)	691		115	
Pre-trend p	0.889		0.576	
Post-period p	<0.001		0.223	

Notes: Both specifications estimated via Double Machine Learning with 3-fold cross-fitting. Placebo assigns pseudo switch dates to non-switchers drawn from the empirical distribution of actual switch dates. Outcome variable is $P(\text{old party} \mid \text{speech})$. Controls include career tenure, legislative activity, legislature fixed effects, and topic fixed effects. Standard errors clustered at deputy level. Pre-trend p from joint Wald test of $\tau = -6$ to -2; Post-period p from joint Wald test of $\tau = +1$ to $+6$. $^{***}p < .001$.

Table 3: Event Study: Effect Size Summary

Estimand	Definition	Value	Source
Event Study (Peak)	τ=+1 vs. τ=-1, covariate-adjusted	−0.0387	Table 2
Window ATE	Mean(all post) − Mean(all pre), unadjusted	−0.0412	Table 14
Baseline $P(\text{old}\mid\text{speech})$	Mean in reference period ($\tau = -1$)	0.1127	—
Covariate-adjusted peak as % of baseline		−34.3%	

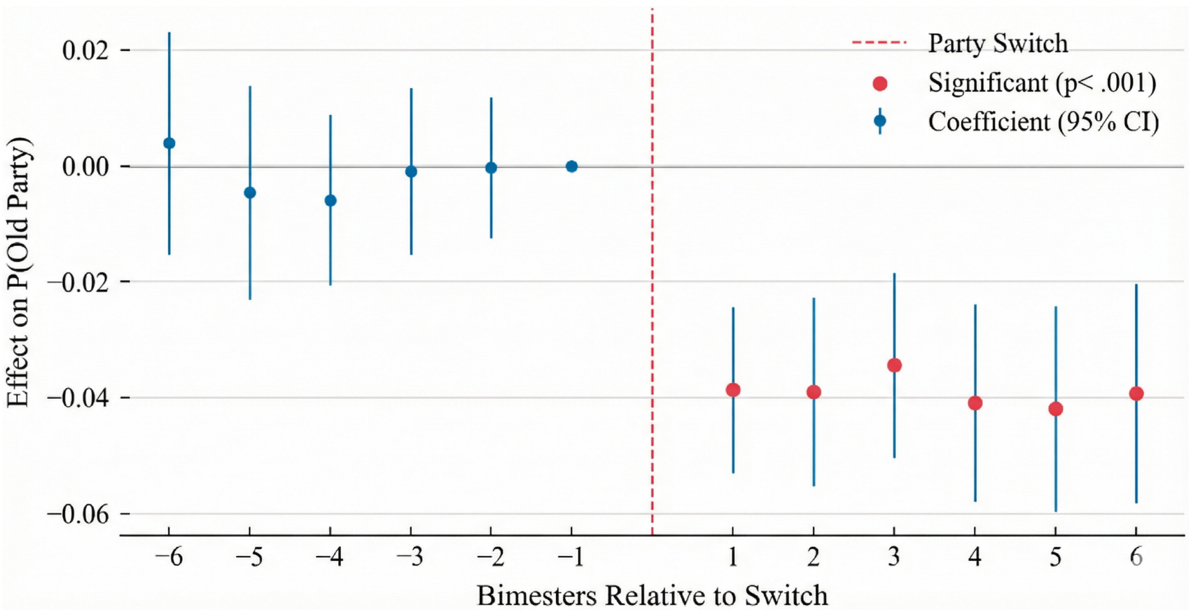

Figure 1: Event Study: Linguistic Adaptation After Party Switching

Notes: Plot shows DML event study coefficients with 95% confidence intervals. X-axis measures bimesters relative to the switch date (0 = switch date). Y-axis measures the change in $P(\text{old party}\mid\text{speech})$ relative to the reference period $\tau = -1$. Negative coefficients indicate that speeches are less likely to be classified as the deputy's old party, consistent with linguistic adaptation. The dashed vertical line marks the switch date. Pre-period coefficients cluster around zero, supporting the parallel trends assumption. Post-period coefficients are negative and significant, indicating substantial adaptation.

6.2 Effect Decomposition

We decompose the adaptation effect into two components: changes in political references (named entities) and changes in policy vocabulary. To do so, we re-estimate the event study using text from which named entities have been removed. The difference between the main specification and the named-entity-removed specification provides an estimate of how much of the detected change reflects shifts in whom deputies reference versus how they discuss policy.

Table 4 presents the decomposition results. Using the main text representation (Level 2), which retains named entities, the peak effect at $\tau = +1$ is -0.0387. Using text with named entities removed (Level 3), the peak effect attenuates to -0.0103. The difference indicates that named entities account for approximately 73% of the total adaptation signal. The remaining 27% reflects changes in policy vocabulary after removing all named entities.

Table 4: Event Study: Effect Decomposition

Text Representation	Peak Effect ($\tau = +1$)	Share of Total
Level 2 (with named entities)	-0.0387^{***}	100%
Level 3 (named entities removed)	-0.0103^{***}	26.6%
Decomposition:		
Named entity component	-0.0284	73.4%
Policy vocabulary component	-0.0103	26.6%

Notes: Level 2 text is normalized with named entities retained. Level 3 text has named entities (persons, organizations, locations) removed using spaCy's Portuguese NER model. Both specifications use Double Machine Learning with 3-fold cross-fitting. Controls include career tenure, legislative activity, legislature fixed effects, and topic fixed effects. Standard errors clustered at deputy level. $N = 47,019$ speech-events. $^{***}p < .001$.

Both components are statistically significant, indicating that deputies change both whom they reference and how they discuss substantive policy after switching parties. However, the named entity component is substantially larger. This pattern is consistent with the interpretation that party switching produces immediate changes in political networks and alliances (deputies begin mentioning their new party's leaders, allies, and affiliated organizations) while changes in policy framing are smaller though still detectable.

The decomposition has implications for interpreting the main result. The adaptation we detect is not primarily ideological repositioning or changes in policy positions. Rather, it reflects a reconfiguration of political references that accompanies the formal change in party affiliation. Deputies quickly adopt the rhetorical practice of mentioning new allies and ceasing to mention former allies, while the underlying policy vocabulary shifts more modestly.

6.3 Voting Behavior

We examine whether voting behavior changes following party switches and whether linguistic and voting changes are related at the individual level. This analysis is limited by sample selection: only 19.4% of switchers have sufficient speech and voting data to compute both measures, and these deputies differ systematically from excluded switchers.

6.3.1 Aggregate Voting Changes

Table 5 presents voting loyalty before and after party switches for the 321 switchers with sufficient voting data.

Table 5: Voting: Average Party Loyalty After Switch

Measure	Mean	Std. Dev.	N
Pre-switch (180 days before):			
Loyalty to old party	0.837	0.196	321
Post-switch (180 days after):			
Loyalty to new party	0.866	0.209	321
Loyalty to old party	0.823	0.220	321
Change:			
New party post − old party pre	+0.029	0.277	321
t = +1.869, p = 0.063			

Notes: Loyalty is the share of roll-call votes aligned with the party majority position. Sample restricted to switchers with at least 3 votes in each window.

Before switching, deputies vote with their old party 83.7% of the time. After switching, they vote with their new party 86.6% of the time, a 2.9 percentage point increase. This difference is marginally significant (t = 1.869, p = 0.063).

6.3.2 Individual-Level Correlation

We test whether deputies who change their rhetoric more also change their voting more. For each switcher, we compute the change in old-party classification probability (pre minus post) and the change in party loyalty (new party post minus old party pre). A positive correlation would indicate that both forms of change tend to occur together.

This analysis requires switchers with at least one speech and at least three votes in each 180-day window, yielding 247 switch events involving 181 unique deputies. This represents 19.4% of switchers. Table 6 shows that included deputies are more active, make larger ideological jumps, and switch more frequently than excluded deputies.

Table 6: Voting: Individual Correlation Sample Selection

Variable	Included (N = 181)	Excluded (N = 752)	p-value
Career tenure (years)	6.82	6.52	0.623
Speeches per year	33.8	22.3	0.000
Total speeches	305.5	173.9	0.001
Total votes	318.5	245.4	0.000
Ideological distance	0.349	0.211	0.000
Number of switches	2.51	1.58	0.000
Switch to major party (%)	20.4	6.1	0.000
Rightward switch (%)	53.6	37.1	0.000

Notes: Included deputies have at least 1 speech and at least 3 votes in each 180-day window. Continuous variables compared using Welch's t-test; categorical variables using χ^2 test.

Table 7 presents the relationship between linguistic and voting adaptation, estimated via OLS regression of voting change on linguistic change. Given the substantial differences between included and excluded switchers, we assess robustness by adding controls for the selection-related characteristics identified in Table 6.

Table 7: Voting: Individual Correlation Results

Model	β (Language)	SE	p-value	N
(1) Baseline	−0.434	0.319	0.174	247
(2) + Selection controls	−0.453	0.331	0.170	243
(3) + Career tenure	−0.438	0.335	0.191	243

Notes: Dependent variable is ΔVoting (loyalty to new party post-switch minus loyalty to old party pre-switch). ΔLanguage is the change in old-party classification probability (pre minus post). Selection controls include ideological distance, number of switches, switch to major party indicator, rightward switch indicator, and speeches per year. Heteroskedasticity-robust standard errors.

The baseline coefficient is negative but not statistically significant. The coefficient remains stable when controlling for observable selection characteristics, changing less than 5%

with the addition of controls.

6.3.3 Interpretation

Three conclusions follow from this analysis. First, we find no evidence of a positive correlation between linguistic and voting change among the 247 switchers with sufficient data; the confidence interval excludes correlations above 0.25. Second, we cannot determine whether the true correlation is zero, weakly negative, or weakly positive but attenuated by measurement error. The test has adequate power to detect standardized effects of $|\beta| \geq 0.20$ but is underpowered for smaller effects (Appendix A.6). Third, this analysis covers only 19.4% of switchers, who differ from excluded deputies on multiple characteristics. Whether these findings generalize to less active switchers is unknown.

7 Robustness Checks

This section examines the stability of our main findings across alternative specifications, measurement approaches, and sample definitions. We assess robustness along the dimensions: estimation methods, classifier performance, ideological bloc classification, permutation tests, window specifications, bin sizes, and an alternative outcome variable.

7.1 Alternative Estimators

The main specification uses Double Machine Learning with legislature fixed effects and topic controls. We compare results to OLS with linear controls and two-way fixed effects (TWFE) with deputy and calendar-month fixed effects.

Table 8: Robustness: Alternative Specifications

Period	OLS	TWFE	DML
$\tau = -6$	+0.0076 (0.0077)	−0.0082 (0.0098)	+0.0039 (0.0098)
$\tau = -5$	−0.0019 (0.0077)	−0.0129 (0.0083)	−0.0046 (0.0094)
$\tau = -4$	+0.0002 (0.0070)	−0.0081 (0.0078)	−0.0059 (0.0075)
$\tau = -3$	+0.0054 (0.0062)	−0.0067 (0.0059)	−0.0009 (0.0074)
$\tau = -2$	−0.0030 (0.0059)	−0.0079 (0.0062)	−0.0003 (0.0062)
$\tau = -1$		(reference period)	
$\tau = +1$	−0.0420*** (0.0079)	−0.0419*** (0.0062)	−0.0387*** (0.0073)
$\tau = +2$	−0.0425*** (0.0081)	−0.0386*** (0.0073)	−0.0390*** (0.0083)
$\tau = +3$	−0.0377*** (0.0089)	−0.0368*** (0.0101)	−0.0344*** (0.0082)
$\tau = +4$	−0.0437*** (0.0087)	−0.0358** (0.0123)	−0.0409*** (0.0087)
$\tau = +5$	−0.0442*** (0.0092)	−0.0362** (0.0123)	−0.0419*** (0.0090)
$\tau = +6$	−0.0357*** (0.0092)	−0.0350** (0.0130)	−0.0393*** (0.0097)
R^2	0.026	0.159	0.012
Pre-trends p	0.439	0.600	0.889
N	47,019	47,019	47,019

Notes: Standard errors clustered by deputy in parentheses. OLS includes career tenure, legislative activity, legislature fixed effects, and topic fixed effects. TWFE includes deputy and calendar-month fixed effects. DML uses 3-fold cross-fitting with the same covariates as OLS. Pre-trends p-value from F-test of joint significance of pre-period coefficients ($\tau = -6$ to $\tau = -2$). $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

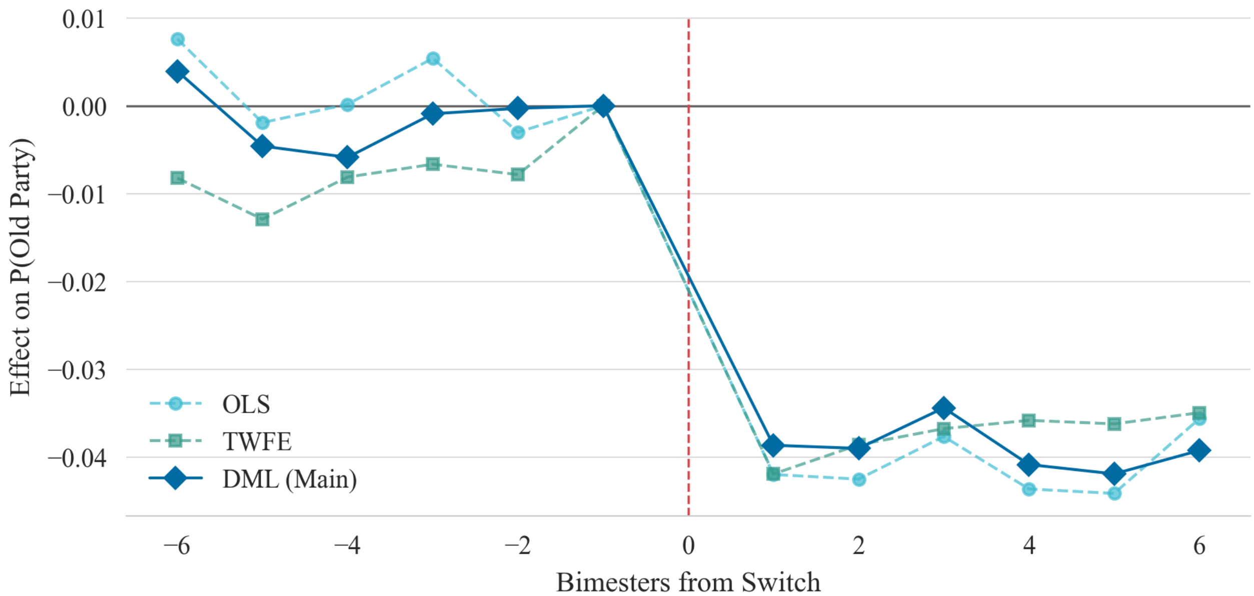

Figure 2: Alternative Estimation Methods

Point estimates are similar across methods. All three specifications yield peak effects between -0.039 and -0.042 at $\tau = +1$. The coefficient correlations across all eleven time periods exceed $r = 0.98$ for every pairwise comparison: OLS and DML ($r = 0.990$), TWFE and DML ($r = 0.979$), and OLS and TWFE ($r = 0.984$).

All three methods pass the parallel trends test: F-statistics for joint significance of pre-period coefficients yield p-values of 0.439 (OLS), 0.600 (TWFE), and 0.889 (DML). Pre-period coefficients are individually insignificant across specifications.

Post-switch coefficients are negative and statistically significant across all methods, with effects persisting through $\tau = +6$.

The R^2 values differ across specifications (0.026 for OLS, 0.159 for TWFE, 0.012 for DML) because TWFE absorbs deputy-level heterogeneity through fixed effects. This affects explained variance but not treatment effect estimates.

7.2 Classifier Performance

7.2.1 Overall Performance

Table 9: Robustness: Classifier Performance

Metric	Value
Overall accuracy	28.9%
Weighted F1 score	0.326
Performance by party size:	
Major parties (top 3)	F1 = 0.349
Medium parties (ranks 4–10)	F1 = 0.212
Minor parties (ranks 11+)	F1 = 0.070

Notes: Metrics from 5-fold cross-validation on the training set (139,677 speeches from 611 non-switchers). Baseline is uniform random guessing across 25 parties (4.0%). Weighted F1 weights each party by frequency.

The classifier achieves 28.9% accuracy, 7.23 times the 4.0% random baseline. The classifier correctly identifies the speaker's party for approximately one-quarter of speeches. However, our outcome variable is the continuous probability $P(\text{old party}|\text{speech})$, not the discrete prediction. Even when the top prediction is incorrect, the probability assigned to the true party contains information. Classification errors occur predominantly within ideological blocs: PT speeches are more often misclassified as PCdoB (both left-wing) than as PSDB (center-right).

7.2.2 Algorithm Selection

Table 10: Robustness: Alternative Classifiers

Algorithm	Accuracy	Weighted F1
Multinomial Naive Bayes	41.0%	0.293
Linear SVM	31.3%	0.344
Logistic Regression	28.9%	0.326
Random Forest	20.8%	0.238

Notes: All classifiers trained on identical TF-IDF features with 5-fold cross-validation. Logistic regression uses class-balanced weights.

We use logistic regression rather than algorithms with higher raw accuracy. Multinomial Naive Bayes achieves 41.0% accuracy but lower weighted F1 (0.293 vs. 0.326). This reflects class imbalance: Naive Bayes concentrates predictions on high-frequency parties, achieving higher overall accuracy but performing poorly on minority parties. Logistic regression with class balancing achieves more uniform performance across parties. Additionally, Naive Bayes produces miscalibrated probability estimates due to violated independence assumptions.

7.2.3 Robustness: Classifier Performance by Party

Table 11: Robustness: Classifier Performance by Party

Party	Speeches	Precision	Recall	F1
PT (Workers' Party)	57,189	0.704	0.305	0.425
PSDB (Brazilian Social Democracy)	21,872	0.307	0.209	0.248
PCdoB (Communist Party)	12,217	0.358	0.391	0.374
PSB (Brazilian Socialist Party)	10,310	0.360	0.317	0.337
PMDB (Brazilian Democratic Movement)	7,675	0.190	0.254	0.217
PDT (Democratic Labor Party)	6,872	0.378	0.404	0.391
PTB (Brazilian Labor Party)	3,948	0.118	0.252	0.160
NOVO (New Party)	2,861	0.119	0.198	0.149
PFL (Liberal Front Party)	2,408	0.034	0.101	0.051
PV (Green Party)	2,308	0.122	0.325	0.177

Notes: Top 10 parties by training set size. Metrics from 5-fold cross-validation.

Performance varies by party size. Major parties achieve higher F1 scores due to more training data and more distinctive profiles.

7.2.4 Qualitative Validation

Table 25 in the Appendix presents example speeches from two switchers, illustrating that the classifier detects substantive rhetorical differences beyond party name mentions. In both cases, pre-switch and post-switch speeches differ in framing and vocabulary despite neither prominently featuring party names.

7.2.5 Ideological Bloc Classification

The party-level classifier might detect party-specific vocabulary rather than ideological content. We train an alternative classifier predicting ideological bloc (Left, Center, Right) rather than party.

Table 12: Robustness: Bloc-Level Classification

Classifier	Classes	Accuracy	Effect	N
Party-level	25 parties	28.9%	−0.0387***	47,019
Bloc-level	3 blocs	55.0%	−0.0094***	41,499

Notes: Bloc classifier trained on Left/Center/Right categories from Power and Zucco scores. Reduced sample reflects switches with valid ideology data for both parties. *** $p < 0.001$.

The bloc classifier achieves 55.0% accuracy (versus 33.3% random baseline) and detects significant adaptation (−0.0094, $p < 0.001$). The smaller magnitude compared to party-level analysis reflects coarser measurement.

7.3 Placebo Test

Section 6.1 presented the main placebo results using DML, showing that non-switchers assigned pseudo switch dates exhibit no systematic change in old-party classification probability. Here we extend that analysis by estimating the placebo test across all three estimation methods (OLS, TWFE, and DML) to verify that the null placebo finding is robust to specification choice.

We randomly assign each of 115 non-switching deputies a pseudo switch date drawn from the empirical distribution of actual switch dates, constrained so that each deputy has speeches in the ±360-day window. We then estimate the same event study specifications used for actual switchers on these placebo events.

Table 13: Robustness: Alternative Specifications for Placebo Test

	Placebo (Non-Switchers)			**Actual Switchers**
	OLS	TWFE	DML	DML
$\tau = +1$ coefficient	+0.0116	+0.0048	+0.0199	−0.0387***
Standard error	(0.0101)	(0.0132)	(0.0112)	(0.0073)
p-value	0.252	0.715	0.075	<0.001
Pre-trends p	0.405	0.061	0.576	0.889
Post-period p	0.366	0.325	0.223	<0.001
N (speech-events)	8,751	8,751	8,751	47,019
N (deputies)	115	115	115	691

Notes: Placebo assigns pseudo switch dates to non-switchers drawn from the empirical distribution of actual switch dates. OLS includes career tenure, legislative activity, legislature fixed effects, and topic fixed effects. TWFE adds deputy and calendar-month fixed effects. DML uses 3-fold cross-fitting. Standard errors clustered at deputy level. Pre-trends p from joint test of $\tau = -6$ to -2; Post-period p from joint test of $\tau = +1$ to $+6$. $^{***}p < .001$.

The placebo test yields null effects across all three specifications. Non-switchers show no systematic change in old-party classification probability around their pseudo switch dates. The $\tau = +1$ coefficients range from +0.0048 to +0.0199, all positive (opposite in sign to the actual effect), small in magnitude, and statistically insignificant. Joint tests for all post-period coefficients are likewise insignificant across specifications (p-values from 0.223 to 0.366).

In contrast, actual switchers exhibit a large negative effect (−0.0387, $p < 0.001$) with highly significant joint post-period effects. The difference between actual and placebo effects is stark: actual switchers show a 34% decline in old-party classification probability, while non-switchers show no detectable change. This supports that our main findings reflect genuine behavioral adaptation to party switching rather than classifier artifacts, temporal drift in speech patterns, or other confounds unrelated to actual switching events.

7.4 Window Specifications

Our main analysis uses a ±12-month window around switch events with 60-day bins. These choices involve tradeoffs: wider windows provide more data but risk contamination from other events; finer bins offer temporal precision but reduce observations per bin. We assess sensitivity to both choices.

7.4.1 Window Width

Wider windows include more speeches but may capture confounding events (e.g., elections, legislative term transitions, or subsequent switches for multiple switchers). Narrower windows reduce sample size but focus on the immediate post-switch period. Table 14 presents average treatment effects across three window specifications.

Table 14: Robustness: Sensitivity to Window Width

Window	ATE	Std. Error	N
±6 months	−0.0412***	(0.0020)	25,052
±9 months	−0.0416***	(0.0016)	36,954
±12 months	−0.0412***	(0.0014)	47,019

Notes: ATE is mean post-switch minus mean pre-switch difference in $P(\text{old party}|\text{speech})$. Standard errors from Welch t-test. *** $p < 0.001$.

Effects are stable across window widths, ranging from −0.0412 to −0.0416. The near-identical point estimates across specifications suggest that adaptation occurs rapidly and persists: the effect detected in the narrow ±6-month window is indistinguishable from that in the ±12-month window. This stability also indicates that our findings are not driven by speeches far from the switch date, which might reflect other political changes rather than party switching per se.

7.4.2 Bin Size

The choice of bin size affects both statistical power (larger bins have more observations) and temporal resolution (smaller bins can detect faster dynamics). Table 15 presents peak effects and parallel trends tests across three bin specifications.

Peak effects range from −0.0325 to −0.0384 across bin sizes, all highly significant. The 30-day specification shows a slightly smaller peak, likely because the first 30-day bin captures a mix of immediate adaptation and any measurement noise around the exact switch date. All three specifications pass the parallel trends test (p-values from 0.275 to 0.499), confirming that the null pre-trend finding is not an artifact of temporal aggregation. The consistency across bin sizes reinforces our conclusion that adaptation begins immediately following the switch.

Table 15: Robustness: Sensitivity to Bin Size

Bin Size	Peak ($\tau = +1$)	Std. Error	Pre-trends p	Periods
30 days	-0.0325^{***}	(0.0069)	0.279	24
60 days	-0.0384^{***}	(0.0089)	0.275	12
90 days	-0.0351^{***}	(0.0079)	0.499	8

Notes: N = 46,987. Peak effect is the $\tau = +1$ coefficient from DML event study. Pre-trends p from joint F-test of all pre-switch coefficients. *** $p < 0.001$.

7.5 Alternative Outcome

Our main outcome, $P(\text{old party}|\text{speech})$, measures how much deputies diverge from their former party's rhetorical patterns. This captures one dimension of adaptation but does not directly assess whether deputies move toward their new party. A deputy could become less similar to their old party without becoming more similar to their new party, for example by adopting idiosyncratic speech patterns or moving toward a third party's rhetoric.

To address this, we construct an alternative outcome measuring the net shift in partisan classification: $\Delta P = P(\text{new party}|\text{speech}) - P(\text{old party}|\text{speech})$. Positive values indicate that speeches are more likely to be classified as the new party than the old party. This outcome requires valid new-party identification in the classifier, reducing the sample size.

Table 16: Robustness: Alternative Outcome

Outcome	Effect	N
P(old party) decline	-0.0387^{***}	47,019
$P(\text{new}) - P(\text{old})$ increase	$+0.0464^{***}$	35,961

Notes: Alternative outcome requires valid new-party identification in the classifier, reducing N. Effects estimated via DML with 3-fold cross-fitting. *** $p < 0.001$.

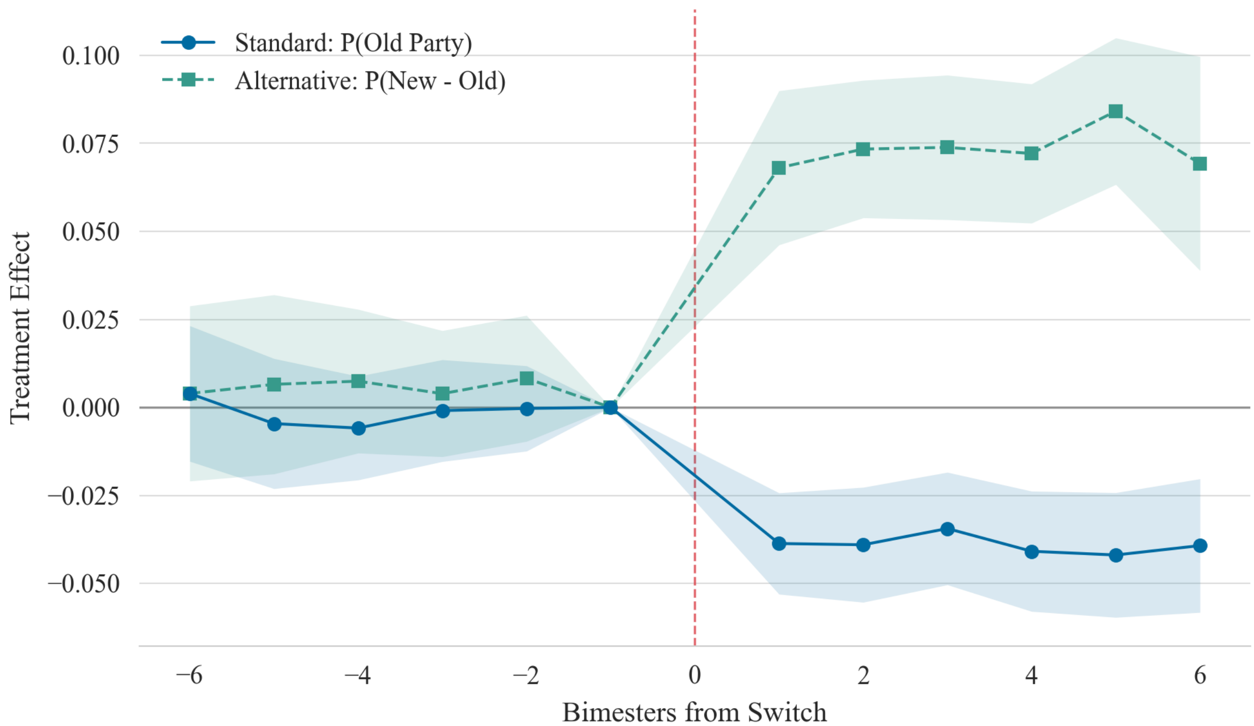

Figure 3: Event Study with Alternative Outcomes. Line shows decline in old-party classification probability (main outcome). Dashed shows increase in the difference between new-party and old-party classification probabilities (alternative outcome). Both outcomes exhibit flat pre-trends and sharp post-switch changes.

The net shift is +0.0464, indicating that speeches become substantially more likely to be classified as the new party relative to the old party after switching. The alternative outcome effect (+0.0464) is slightly larger in magnitude than the main outcome effect (−0.0387), suggesting that deputies do not merely move away from their old party but actively move toward their new party's rhetorical patterns. Figure 3 shows that both outcomes exhibit flat pre-trends and sharp discontinuities at the switch date, confirming that the adaptation we detect reflects genuine movement toward the new party rather than generic divergence from the old party.

7.6 Summary

wThe main findings are robust across specifications, measurement approaches, and sample definitions. Table 17 summarizes the robustness checks.

Table 17: Summary of Robustness Checks

Check	Variation	Result
Estimation method	OLS, TWFE, DML	Effects stable (-0.039 to -0.042)
Classifier algorithm	NB, SVM, LR, RF	LR selected for calibration
Outcome level	Party vs. bloc	Both significant; party more precise
Placebo test	Non-switchers	Null effects across methods
Window width	± 6, 9, 12 months	Effects stable (-0.041 to -0.042)
Bin size	30, 60, 90 days	Effects stable (-0.033 to -0.038)
Alternative outcome	$P(\text{new}) - P(\text{old})$	Confirms movement toward new party

Three findings emerge. First, the magnitude and significance of rhetorical adaptation are insensitive to estimation method: OLS, TWFE, and DML yield nearly identical coefficients with correlations exceeding $r = 0.98$. Second, the effect reflects genuine behavioral change rather than statistical artifacts: placebo tests on non-switchers yield null effects, and parallel trends hold across all specifications. Third, deputies move toward their new party's rhetoric, not merely away from their old party: the net classification shift (+0.046) exceeds the old-party decline (-0.039) in magnitude.

8 Heterogeneity Analysis

This section examines whether linguistic adaptation varies across observable characteristics. We focus on four dimensions: career experience, ideological distance, switch direction, and destination party size.

These analyses are exploratory. We did not pre-register predictions, and we examine multiple dimensions, raising concerns about multiple comparisons. We interpret the results as descriptive patterns rather than tests of specific hypotheses. All tests have adequate power to detect medium effects; Appendix A.6.2 reports full power calculations.

Heterogeneity analysis uses switcher-level effects computed as the pre minus post difference in $P(\text{old party} \mid \text{speech})$ for each deputy with sufficient speeches in both windows (659 switchers).

8.1 Ideological Distance

We regress the absolute magnitude of linguistic adaptation on continuous ideological distance, measured as $|\text{EST}_{\text{new}} - \text{EST}_{\text{old}}|$ on the Power and Zucco scale.

Table 18: Heterogeneity by Ideological Distance

Variable	Coefficient
Ideological distance	-0.0044
Constant	0.0727***
N = 519, $R^2 = 0.001$	
Pearson $r = -0.024$, Spearman $\rho = -0.038$	

The relationship is close to zero. The regression coefficient is small, negative, and not statistically significant ($p = 0.591$). Deputies who traverse larger ideological distances do not show larger adaptation effects than those making smaller moves.

8.2 Career Experience

We compare adaptation across three experience categories based on years since first election to the Chamber.

Table 19: Heterogeneity by Career Experience

Experience Group	Mean $\|\Delta\text{Language}\|$	SE	N
Junior (<5 years)	0.0621	(0.0044)	297
Mid-career (5–10 years)	0.0578	(0.0057)	151
Senior (>10 years)	0.0531	(0.0042)	211
F(2, 656) = 0.994, p = 0.370			

The differences across experience groups are not statistically significant ($p = 0.370$). Junior deputies show the highest mean adaptation (0.0621), approximately 17% higher than seniors (0.0531), but the difference does not reach significance.

8.3 Switch Direction

We compare adaptation by direction on the left-right spectrum.

Table 20: Heterogeneity by Switch Direction

Direction	Mean $\|\Delta\text{Language}\|$	SE	N
Rightward	0.0715	(0.0088)	135
Leftward	0.0447	(0.0048)	83
Lateral (within-bloc)	0.0576	(0.0038)	301
t-test (Rightward vs. Leftward): t = 2.265, p = 0.025			

Rightward and leftward switches differ significantly. Deputies switching rightward show mean adaptation of 0.0715, while those switching leftward show 0.0447, a 60% difference (t = 2.265, p = 0.025).

This finding has multiple possible interpretations. Right-wing parties may have more distinctive rhetorical profiles, making adaptation more detectable when deputies join them. Alternatively, deputies who switch rightward may differ from those switching leftward on unobserved characteristics. The asymmetry survives controls for ideological distance, career tenure, and destination party size, but we cannot determine which explanation accounts for the pattern.

8.4 Destination Party Size

We examine whether adaptation varies by destination party size.

Table 21: Heterogeneity by Destination Party Size

Party Size	Mean $\|\Delta\text{Language}\|$	SE	N
Major (≥30 deputies)	0.0564	(0.0033)	360
Medium (10–29 deputies)	0.0514	(0.0042)	170
Minor (<10 deputies)	0.0724	(0.0089)	129
F(2, 656) = 3.602, p = 0.028			

Deputies joining minor parties show larger adaptation (0.0724) than those joining major parties (0.0564), a 28% difference (p = 0.028). However, we interpret this finding with considerable caution due to a potential measurement artifact. Our classifier is trained on substantially more speeches from major parties than minor parties (Table 11), and classifier performance varies accordingly: major parties achieve F1 scores of 0.349 compared to 0.070 for minor parties. This asymmetry means the classifier is effectively a better detector of major-party rhetoric than minor-party rhetoric. When a deputy switches from a major party to a minor party, the classifier may register a large decline in old-party probability simply because it could identify the old party well but cannot identify the new party well, a mechanical relationship that mimics adaptation. Conversely, switches to major parties may show smaller measured adaptation because the classifier assigns non-trivial probability to the destination party even before the switch. We therefore do not emphasize this result as substantively informative about differential adaptation by party size, as we cannot separate genuine behavioral differences from classifier performance differences.

8.5 Summary

Table 22: Summary of Heterogeneity Checks

Dimension	Finding	p-value
Ideological distance	No relationship ($r = -0.02$)	0.591
Career experience	No significant difference	0.370
Switch direction	Rightward > Leftward	0.025
Destination party size	Minor > Major	0.028

Notes: Power analysis in Appendix A.6.2.

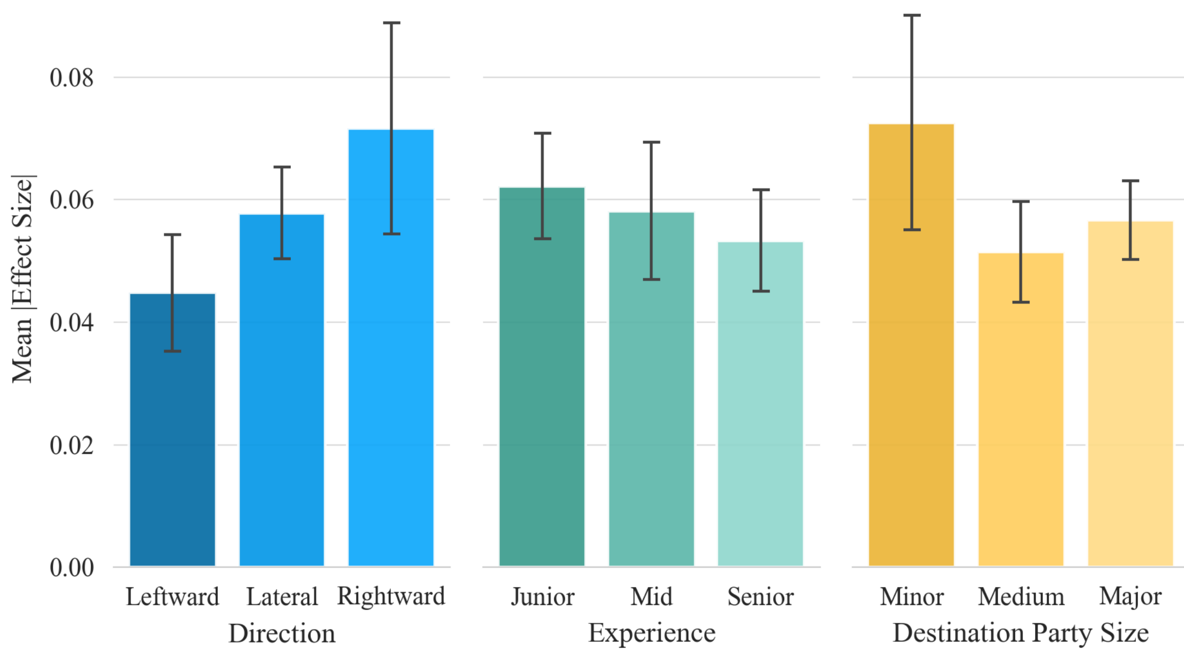

Figure 4: Heterogeneity by Direction, Experience, and Party Size

Two dimensions show no significant variation (ideological distance, career experience). Two dimensions show significant variation: rightward switches produce larger effects than leftward switches, and switches to minor parties produce larger effects than switches to major parties. The null results indicate that adaptation does not scale with the magnitude of repositioning or vary systematically with seniority. The significant findings suggest that political context matters, though alternative explanations involving classifier performance or selection cannot be excluded for either result.

9 Discussion

This study examined linguistic change following party switches, analyzing 365,551 speeches from Brazil's Chamber of Deputies. We document substantial rhetorical adaptation after switching, decompose its sources, and assess how linguistic change relates to voting behavior.

9.1 Findings and Interpretation

Deputies' speeches shift away from their former party's linguistic patterns within one bimester of switching. The probability that a speech is classified as belonging to the old party declines by 3.9 percentage points, corresponding to a 34% reduction from the pre-switch baseline. This effect emerges quickly and persists throughout the observation window, indicating that rhetorical adaptation is both immediate and durable.

A decomposition of the linguistic signal shows that adaptation operates through two distinct channels. Approximately 73% of the change reflects shifts in political references, including the politicians, organizations, and locations that deputies mention. The remaining 27% reflects changes in policy-related vocabulary after removing named entities. Both components are statistically significant, but the reference component is substantially larger. This pattern suggests that adaptation is driven primarily by changes in political networks and focal actors, rather than by deep changes in policy framing.

At the aggregate level, party switching is associated with a marginally significant increase in voting loyalty to the new party of 2.9 percentage points ($p = 0.063$). However, when focusing on the 247 switchers with sufficient data to compute individual-level measures of both rhetoric and voting, we find no positive correlation between the two domains ($r = -0.114$, $p = 0.074$). The confidence interval excludes correlations larger than 0.25. This divergence between aggregate and individual-level results suggests that while switching is associated with modest average changes in both speech and voting, these changes do not systematically move together within individuals.

Several interpretations are consistent with this pattern. Linguistic adaptation may be driven by social exposure and interaction with new colleagues, which alters whom deputies reference more than how they vote. Voting behavior, in contrast, may be shaped by party discipline, coalition agreements, and strategic considerations that are only weakly related to rhetorical alignment. Measurement error may also play a role. Both linguistic probabilities and voting loyalty are noisy proxies for underlying partisan alignment, which could attenuate correlations even if a weak positive relationship exists. In addition, the individual-level correlation analysis covers only 19.4% of switchers, who are more active than excluded deputies, limiting our ability to assess generalizability.

Heterogeneity analyses show little evidence that adaptation varies with individual characteristics such as career experience ($p = 0.370$) or ideological distance between parties ($p = 0.591$). In contrast, adaptation varies by political context. Rightward switches generate larger linguistic shifts than leftward switches ($p = 0.025$). This asymmetry may reflect differences in party organization, classifier performance across parties, or coalition dynamics during the Temer and Bolsonaro administrations, when rightward switches frequently involved joining the governing coalition.

9.2 Limitations

We acknowledge several limitations.

Sample selection. The rhetoric voting correlation analysis includes only 19.4% of party switchers. Included deputies are more active, switch more frequently, and make larger ideological moves than excluded deputies (Table 6). Although our results are robust to controls for these observable differences, we cannot rule out selection on unobservables. Whether the absence of a positive rhetoric voting correlation extends to less active switchers remains unknown.

Measurement error. Our classifier achieves 28.9% accuracy, well above the 4% random baseline but still leaves most speeches misclassified. The continuous classification probabilities we use are therefore noisy measures of partisan rhetorical alignment. Measurement error in both linguistic and voting measures may attenuate correlations toward zero, preventing us from distinguishing a true null relationship from a weak positive association masked by noise.

External validity. Brazil's party system is unusually fluid, with 25 parties in our sample and frequent switching. The results may not generalize to systems with stronger party institutionalization, where switching is rarer and more costly. The mechanisms identified here, particularly network realignment and limited integration of policy language, may operate differently where party brands are more stable and salient to voters.

9.3 Conclusion

Party switching in Brazil leads to rapid and persistent linguistic adaptation in legislative speech. Most of this adaptation reflects changes in political references rather than shifts in policy vocabulary, highlighting the central role of social and organizational alignment. While party switching is associated with a modest increase in voting loyalty at the aggregate level, individual-level changes in rhetoric and voting are not positively correlated among deputies with sufficient data. Together, these findings indicate that rhetorical and behavioral alignment represent distinct dimensions of partisan adaptation, shaped by different constraints and incentives within the legislative environment.

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A Appendix

A.1 Descriptive Statistics

Table 23: Appendix: Distributional Characteristics

Variable	**Mean**	**Std. Dev.**
Corpus		
Speeches per deputy	224	448
Words per speech (Level 2)	112	151
Deputies		
Career tenure (years)	6.6	7.3
Legislative activity (annual)	114.7	152.6
Switching		
Switches per switcher	1.76	1.13
Ideological distance	0.259	0.291

Notes: The corpus covers legislatures 52–58 (2003–2025). Level 2 preprocessing includes normalization and stopword removal while retaining named entities. Ideological distance is $|EST_{new} - EST_{old}|$ on the Power and Zucco scale.

Table 24: Appendix: Sample Sizes

Category	**Count**
Full corpus	
Total speeches	365,551
Total deputies	1,634
Total roll-call votes	388,911
Party switching	
Switch events	1,642
Unique switchers	933
Multiple switchers (>1 switch)	406
Ideology (switches with valid data)	
Switches with valid ideology	1,101
Within-bloc switches	713
Cross-bloc switches	388
Analysis samples	
Event study speech-events	47,019
Event study unique switchers	691
Switchers with voting data	321
Correlation sample	247

Notes: Within-bloc switches are those where origin and destination parties share the same ideological category (Left, Center, or Right). The correlation sample requires at least 1 speech and at least 3 votes in each 180-day window.

A.2 Appendix: Sample Construction

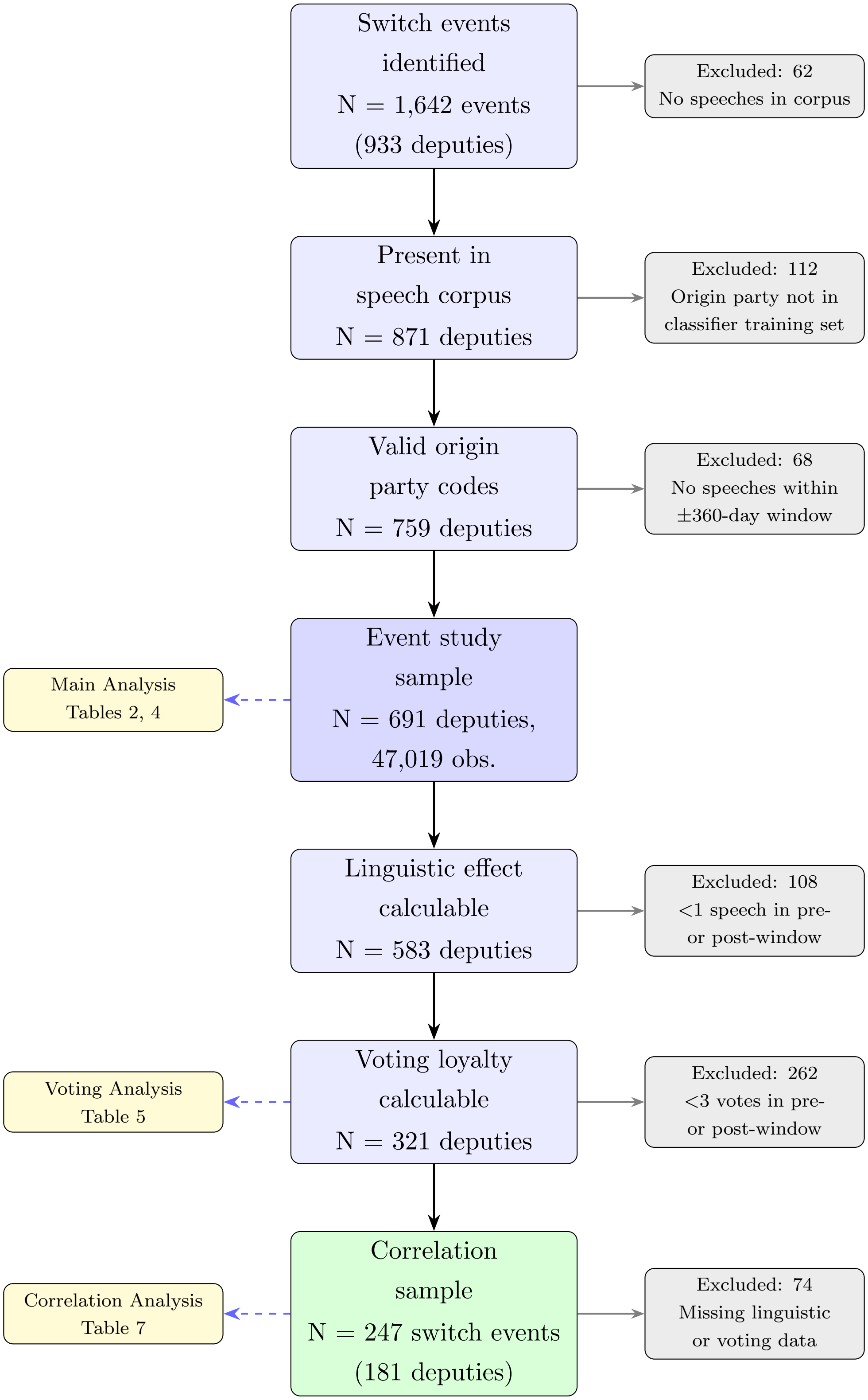

Figure 5: Sample Construction

A.3 Classifier Samples

Table 25: Appendix: Example Speeches

Example 1: PSOL to PSB (Left to Left)		
Pre-switch	0.998	"We have a President who denied isolation, a President who didn't wear a mask. We have a President who denied science and spoke against the vaccine. We have a country with more than 170,000 Brazilians dead. We have a President who called a pandemic that stopped the world a 'little flu.' So it is evident that [we vote] in favor of having more resources for vaccine development and that these resources can be directed to FIOCRUZ, which plays a decisive role in..."
Post-switch	0.003	"Mr. President, before passing the Minority Leadership to my friend Deputy Alencar, I will do so with great honor, I want to speak here about the tragedy that occurred in the municipality of Petrópolis. Petrópolis is one of the most beautiful cities in Brazil. It is an imperial city, a historic city in the mountain region of Rio de Janeiro, with more than 300,000 residents.."
Example 2: PTB to PPS (Right to Left)		
Pre-switch	0.990	"Mr. President, [we] recommend a 'yes' vote. We heard various concepts and reflections in this chamber. I am a doctor and I insist on saying that pregnancy requires the mother's womb. The woman's sanctuary is her womb, and nobody is going to have an abortion to help with research. What will be used are embryos that are frozen and would be destined for the trash. [We vote] 'yes.' (Applause from the galleries.)"
Post-switch	0.006	"Mr. President, I want once again to express my concern about the transposition of waters from the São Francisco River. We understand that first the hydrographic basin must be revitalized, the river of national unity. As a resident of the central-west region of Minas Gerais, through my city, Divinópolis, flow two rivers: Itapecerica and Pará, where the São Francisco is born and from where it flows to all of Brazil, reaching the Atlantic Ocean, I want to say that..."

Notes: Examples illustrate classifier behavior. Example 1 shows a within-left switch: the pre-switch speech uses confrontational framing; the post-switch speech uses neutral language about a natural disaster. Example 2 shows a cross-ideological switch: the pre-switch speech uses religious framing on stem cell research; the post-switch speech discusses environmental policy. Neither speech prominently features party names, yet P(old party) drops dramatically in both.

A.4 Topic Model

Table 26: Appendix: LDA Topics

ID	Top Words (Stemmed)	Interpretation
0	inform, solicit, investig, ped, denúnc, divulg, audi	Information requests
1	transport, águ, indígen, ambient, florest, rodov, estr	Infrastructure/environment
2	indústr, export, vacin, comérci, import, exteri, comerc	Trade/foreign relations
3	impost, fiscal, pag, tributár, receit, tribut, vet	Taxation/fiscal policy
4	vot, entend, aprov, apresent, orient, retir, discut	Legislative procedure
5	negr, comemor, cult, reconhec, viv, marc, histór	Culture/racial issues
6	médic, mort, hospit, agent, doenç, crim, penal	Healthcare/justice
7	realiz, prefeit, municip, particip, event, visit, receb	Local government
8	fal, acontec, viv, cheg, bolsonar, fic, pass	Political discourse
9	produt, rural, agricul, famili, aliment, pequen, agrícol	Agriculture
10	mulh, human, adolesc, igrej, crianç, sex, filh	Social issues/religion
11	pesquis, ger, produt, sustent, desenvolv, tecnolog	Technology/development
12	candidat, corrupç, companh, jorn, democrá, eleit	Elections/corruption
13	dev, cert, mudanç, lev, permit, pod, enfrent	Generic political
14	reg, intern, distrit, parec, destaqu, orçamentár	Budget/regulations
15	milit, arm, polic, drog, forç, sal, pré	Military/security
16	estabelec, apresent, refer, alter, aplic, regulament	Legal frameworks
17	trabalh, previd, aposent, reajust, categor, salári	Labor/social security
18	escol, univers, ensin, profes, estud, alun, curs	Education
19	energ, preç, consum, financ, invest, petrobr, jur	Energy/finance

Notes: LDA with $K = 20$ on 365,551 speeches (Level 4 stemmed text). Top 7 words shown.

A.5 Transition Matrix

Table 27: Appendix: Bloc Transition Matrix

	Destination Bloc			
Origin Bloc	**Center**	**Left**	**Right**	**Total**
Center	131	32	114	277
Left	59	44	78	181
Right	77	28	538	643
Total	267	104	730	1,101

Notes: Switches with valid ideology data for both parties. Diagonal entries are within-bloc switches. Right-to-Right switches account for 48.9% of switches; Left-to-Left for 4.0%.

A.6 Power Analyses

A.6.1 Correlation Power Analysis

A null correlation finding is only informative if we have adequate statistical power to detect correlations that might exist. The regression coefficient reported in Table 7 ($\beta = -0.434$) and the Pearson correlation ($r = -0.114$) test the same hypothesis but on different scales: the regression coefficient is unstandardized (in raw units of voting change per unit of linguistic change), while the correlation is standardized (both variables expressed in standard deviation units). In a bivariate regression, the standardized coefficient equals the Pearson correlation. We conduct power analysis on the correlation scale because effect size benchmarks (e.g., Cohen's conventions) are defined for correlations.

With a sample of N = 247 switch events and a two-tailed test at $\alpha = 0.05$, our ability to detect a true correlation depends on its magnitude.

Table 28: Power Analysis for Correlation Test

True Correlation	Power	Assessment
$\lvert r\rvert = 0.100$	34.7%	Underpowered
$\lvert r\rvert = 0.150$	65.6%	Underpowered
$\lvert r\rvert = 0.200$	88.6%	Adequate
$\lvert r\rvert = 0.250$	97.9%	Adequate
$\lvert r\rvert = 0.300$	99.8%	Adequate
Minimum detectable effect at 80% power: $\lvert r\rvert \geq 0.177$		

Notes: Power is the probability of detecting a true correlation of the given magnitude at $\alpha = 0.05$ (two-tailed). Calculations do not account for attenuation from measurement error; if either measure has reliability below 1, effective power is lower than reported.

The power analysis clarifies what our null finding can and cannot tell us. We have adequate power (88.6%) to detect correlations of $\lvert r\rvert \geq 0.200$, and near-certain power (97.9%) to detect correlations of $\lvert r\rvert \geq 0.250$. Since we do not detect a significant correlation, we can confidently rule out moderate-to-large positive associations between linguistic and voting adaptation.

However, we lack power to detect smaller correlations. With only 34.7% power to detect $\lvert r\rvert = 0.100$, we cannot distinguish a true null from a small positive correlation. A correlation of $r = 0.10$ would explain only 1% of the variance, a substantively trivial relationship, but we cannot exclude it.

Our observed correlation ($r = -0.114$, $p = 0.074$) is not statistically significant at $\alpha = 0.05$. The 95% confidence interval (-0.24 to $+0.01$) includes zero and small negative values but excludes moderate-to-large positive correlations. This is the key result: we cannot determine whether the true correlation is zero, slightly negative, or slightly positive, but we can rule out the strong positive correlation ($r > 0.25$) that a genuine conversion account would predict. Importantly, this conclusion holds whether we examine the bivariate relationship or the regression with selection controls (Table 7), where the coefficient remains negative and insignificant ($\beta = -0.453$, $p = 0.170$).

A.6.2 Heterogeneity Power Analysis

Table 29 reports minimum detectable effects at 80% power for each heterogeneity test.

Table 29: Power Analysis for Heterogeneity Tests

Dimension	**Test**	**N**	**MDE (80%)**	**Power (medium)**
Career experience	ANOVA	659	$f = 0.068$	100%
Switch direction	t-test	218	$d = 0.393$	94.6%
Ideological distance	Correlation	519	$\|r\| = 0.123$	100%

Notes: MDE is minimum detectable effect at 80% power with $\alpha = 0.05$. Power (medium) reports power to detect medium effects by Cohen's conventions ($f = 0.25$ for ANOVA, $d = 0.50$ for t-test, $r = 0.30$ for correlation).

All tests have adequate power to detect medium effects. The ANOVA for career experience and the correlation test for ideological distance both achieve 100% power for medium effects. The t-test for switch direction has 94.6% power for medium effects.